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import tensorflow as tf
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.preprocessing.image import load_img,
img_to_array

#Create an image data generator with augumenntation
datagen = ImageDataGenerator(
    rotation_range = 40,
    width_shift_range = 0.4,
    height_shift_range=0.4,
    shear_range =0.4,
    zoom_range =0.4,
    horizontal_flip =True,
    rescale=None,
    fill_mode='wrap' # constant,wrap,nearest
)

# Load image from path
img = load_img(r"C:\Users\Lenovo\OneDrive\Desktop\Python\VS_Code\Mood
class classification-CNN\training\happy\3.jpeg")

img

```



```

x = img_to_array(img) # this is a Numpy array with shape (3, 150,
150)
x = x.reshape((1,) + x.shape) # this is a Numpy array with shape (1,
3, 150, 150)

```

```
# the .flow() command below generates batches of randomly transformed
images
# and saves the results to the `preview/` directory
i = 0
for batch in datagen.flow(x, batch_size=1,
                        save_to_dir=r"C:\Users\Lenovo\OneDrive\
Desktop\Python\VS_Code\augumented images", save_prefix='cat',
save_format='jpeg'):
    i += 1
    if i > 30:
        break # otherwise the generator would loop indefinitely
```